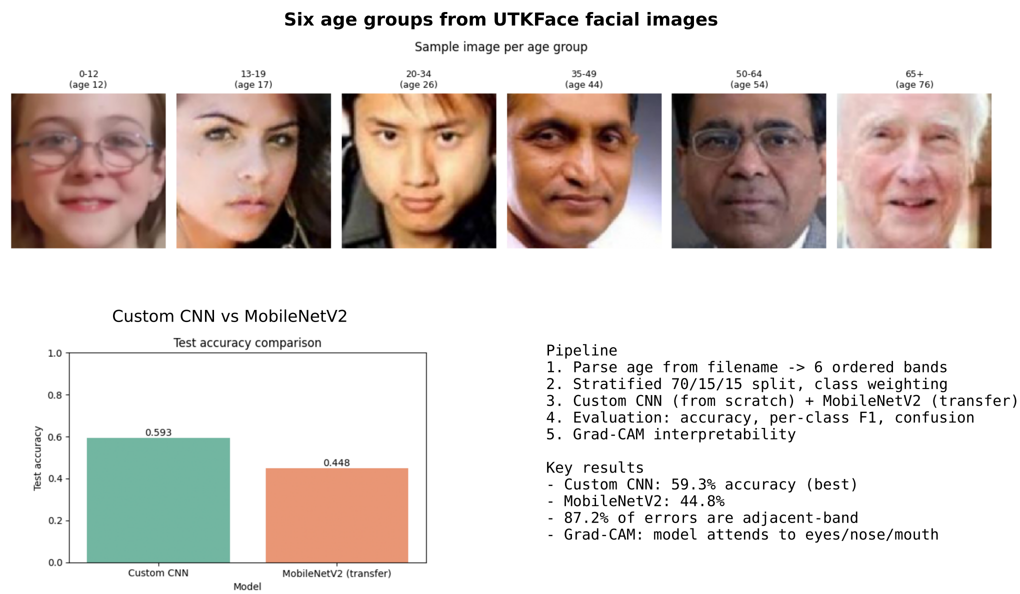
Graphical Abstract

An Explainable Convolutional Neural Network Framework for Six-Class Age-Group Estimation from Facial Images

Pranav Chitirala



Highlights

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- Six-class age-group estimation from 23,708 UTKFace facial images using a custom CNN.
- The from-scratch custom CNN reaches 59.3% test accuracy and a weighted F1-score of 0.60.
- A pretrained MobileNetV2 reached only 44.8%, underperforming the custom baseline.
- 87.2% of all errors fall between adjacent age bands, not distant ones.
- Grad-CAM confirms reliance on the eyes, nose, and mouth regions for age cues.

An Explainable Convolutional Neural Network Framework for Six-Class Age-Group Estimation from Facial Images

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Abstract

Estimating a person's age from a facial image is a long-studied computer-vision problem with applications in access control, audience analytics, content personalisation, and demographic research. Predicting an exact age is brittle, so this study frames the task as the more robust problem of coarse age-group classification, yet most published age-group studies report aggregate accuracy alone and rarely pair it with interpretability or per-class reliability analysis. This work classifies faces into six ordered age bands using the UTKFace dataset and evaluates a custom convolutional neural network against a transfer-learning model not only on predictive accuracy but also on per-class reliability and explainability. A custom convolutional network was trained from scratch on 23,708 UTKFace images, binned into six age groups and split with stratified sampling, using class weighting to counter the strong over-representation of the young-adult band, and was compared against an ImageNet-pretrained MobileNetV2 transfer-learning model. On a held-out test set the custom network attained an overall accuracy of 59.3%, a weighted F1-score of 0.60, and recognised the youngest band most reliably, whereas the MobileNetV2 model reached only 44.8% and was outperformed by the from-scratch baseline. A central finding is that 87.2% of all misclassifications fall between adjacent age bands rather than distant ones, confirming that the residual error is dominated by the genuine visual similarity of neighbouring ages. Grad-CAM analysis shows that predictions are driven by informative facial regions such as the eyes, nose, and mouth, demonstrating that the model relies on meaningful ageing cues rather than background artefacts. The contribution is a reproducible, explainable six-class age-estimation study that compares a custom and a transfer-learning model

and characterises where each succeeds and fails.

Keywords: Age estimation, Convolutional neural networks, Transfer learning, Explainable AI, Facial image classification, Class imbalance

1. Introduction

1.1. Application Background and Practical Importance

The automatic estimation of a person's age from a facial image is a well-established computer-vision task with a wide range of practical applications. Age-aware systems support age-restricted access control, audience analytics in retail and advertising, content personalisation, and large-scale demographic research [1]. The human face carries strong visual cues to age, including skin texture, facial proportions, and the presence of features such as wrinkles, but these cues vary widely across individuals, ethnicity, lighting, pose, and image quality [2]. The people and organisations affected by age-estimation technology include businesses that must verify customer age, platforms that tailor content to audiences, and researchers studying population structure [3]. Manual age assessment by human observers is slow, subjective, and inconsistent, and it does not scale to the volumes of imagery generated by modern systems [4]. Inaccurate or delayed age assessment can lead to failures of access control, mis-targeted content, and biased demographic statistics, all of which carry practical and ethical costs [5]. Automated image analysis is attractive because it can apply consistent criteria at scale, although the inherent ambiguity of facial ageing means that predicting a coarse age band is both more robust and sufficient for many of these uses [6]. This study therefore targets six-class age-group estimation rather than exact-age regression, which is the framing adopted throughout [1].

1.2. Machine Learning and Computer Vision Context

Image-based machine learning has been transformed by convolutional neural networks, which now dominate visual recognition [7]. A convolutional network learns hierarchical visual features directly from pixels, removing the need for the hand-crafted descriptors that earlier age-estimation systems relied upon [5]. Early layers capture low-level cues such as edges and textures, while deeper layers compose these into higher-level structures corresponding to facial parts and ageing patterns [7]. This hierarchical capacity is especially relevant to age estimation, because age is expressed through a combination of fine texture, such as wrinkles, and coarse geometry, such as facial proportion [2]. Convolutional networks are also spatially equivariant, allowing them to detect age-relevant facial regions regardless of small positional shifts [8]. The flexibility of deep networks comes at the cost of a need for large labelled datasets, without which they overfit [9]. For age estimation this is compounded by the natural imbalance of face datasets, in which working-age adults are heavily over-represented [10]. The methodological challenge addressed here is therefore reliable six-class estimation under realistic data and compute constraints [10].

1.3. Transfer Learning for the Selected Problem

Training a deep network from random initialisation is difficult when labelled data are limited, because the model has many parameters and comparatively little signal [11]. Transfer learning mitigates this by reusing a backbone pre-trained on a large generic image collection such as ImageNet [12]. The pre-trained features, including edge, texture, and shape detectors, transfer well to facial analysis tasks including age estimation [13]. Two strategies are common, namely feature extraction with a frozen backbone and fine-tuning of selected backbone layers on the target data [12]. Lightweight backbones such as MobileNetV2 are attractive for age estimation on constrained hardware because of their efficient inverted-residual design [14]. Deeper residual networks such as ResNet offer more capacity through skip connections at higher computational cost [8]. This study trains a custom convolutional baseline and identifies MobileNetV2 as the natural transfer-learning comparator, consistent with the project proposal, so that the accuracy-efficiency trade-off can be examined [15]. A full technical description of each architecture is deferred to the Methodology [15].

1.4. Explainable AI and Trustworthiness

High accuracy alone does not justify trust in a model whose outputs may inform access decisions or demographic conclusions [16]. A model can attain high accuracy while exploiting spurious correlations, such as background or image-source artefacts, that do not reflect genuine ageing cues [17]. Explainable artificial intelligence exposes this risk by indicating which image regions most influence a prediction [18]. Grad-CAM is a gradient-based method that produces class-specific heatmaps highlighting the spatial regions driving an output [19]. For age estimation the expected behaviour is that the model attends to age-informative facial regions such as the eyes, nose, and mouth rather than to the background [2]. Because age-estimation systems can affect access, fairness, and privacy, interpretability is treated here as an evaluation goal rather than an afterthought [16]. This study applies Grad-CAM and assesses whether the highlighted regions are consistent with domain expectations about facial ageing [17].

1.5. Research Gap

A review of the age-estimation literature reveals several recurring limitations. First, many studies report only aggregate accuracy or mean absolute error and provide limited per-class analysis, which obscures poor performance on under-represented age bands [1]. Second, although facial age datasets are strongly imbalanced, explicit imbalance handling is inconsistently applied and reported [10]. Third, interpretability is frequently absent, and where it appears it is usually a small number of qualitative heatmaps without systematic discussion [3]. Fourth, systematic comparison between a custom network and a transfer-learning model under identical conditions is reported less often than it should be [1]. Fifth, statistical reliability and efficiency analysis are reported unevenly, weakening claims of practical applicability [16]. These limitations are not universal, and it would be inaccurate to claim that no study addresses any of them, but their combination in a single evaluation is uncommon [2]. This study responds by pairing six-class estimation with explicit imbalance handling, per-class reliability analysis, a custom-versus-transfer-learning comparison, and Grad-CAM interpretation in one reproducible pipeline [20].

1.6. Problem Statement

This study addresses the problem of automatically classifying a single RGB facial image into one of six ordered age groups using an explainable convolutional-neural-network framework. The input is a cropped facial image and the prediction target is the discrete age band, obtained by binning the continuous age label provided by the dataset. The number of classes is six, and the dominant modelling challenges are the strong imbalance toward the young-adult band and the visual similarity between neighbouring age bands. The study investigates how accurately a custom convolutional network can perform this task, whether transfer learning improves on it, how its errors are structured across age bands, and which facial regions drive its predictions. The practical purpose is a coarse age-estimation component suitable for analytics and access-support applications in which approximate age is sufficient and human oversight is retained.

1.7. Aim and Objectives

The aim of this study is to develop and evaluate an explainable convolutional-neural-network framework for the six-class estimation of age groups from facial images. The specific objectives are as follows.

- To parse and preprocess the UTKFace dataset, including binning continuous ages into six ordered groups.
- To develop a custom convolutional neural network as a reproducible baseline for age-group estimation.
- To apply an explicit class-imbalance treatment and examine its effect on under-represented bands.
- To evaluate predictive performance using accuracy and per-class precision, recall, and F1-score.
- To analyse the structure of errors, in particular whether confusion concentrates between adjacent age bands.
- To explain predictions using Grad-CAM and to assess whether the attended regions are age-informative.
- To compare the custom network against a MobileNetV2 transfer-learning model and to discuss deployment implications.

1.8. Research Questions

The study is organised around the following four research questions taken from the project proposal.

- RQ1: How accurately can a custom CNN classify faces into six age groups from the UTKFace dataset?
- RQ2: Does transfer learning with a pre-trained MobileNetV2 outperform a CNN trained from scratch on this task?
- RQ3: Which age groups are most often confused with one another, and do errors concentrate between adjacent age bands?
- RQ4: Which facial regions does the model rely on when predicting age, as revealed by Grad-CAM?

1.9. Contributions and Novelty

This study makes a set of focused contributions beyond the routine application of convolutional networks.

- A fully reproducible six-class age-estimation pipeline on UTKFace, from filename-based age parsing through stratified splitting to evaluation.
- A custom convolutional baseline trained with class weighting to counter the severe imbalance of the dataset.
- A quantitative error-structure analysis demonstrating that the great majority of misclassifications fall between adjacent age bands.
- A Grad-CAM interpretation assessing whether the model attends to age-informative facial regions.
- A direct comparison showing that a compact custom network outperforms a much larger MobileNetV2 transfer-learning model on this task.

1.10. Report Organization

The remainder of this report is organised as follows. Section 2 reviews related work on age estimation, convolutional and transfer-learning methods, preprocessing, evaluation, and explainability, and presents a comparative literature matrix. Section 3 describes the full methodology, Section 4 reports the results organised by research question, and Section 5 discusses them and states limitations and future work. Section 6 concludes.

2. Literature Review

2.1. Image-Based Machine Learning in the Application Domain

Facial age estimation has evolved through three broad methodological phases. Early approaches relied on hand-crafted geometric and texture descriptors, such as anthropometric ratios and local binary patterns, which were sensitive to pose and illumination [2]. A second phase combined engineered features with classical regressors and classifiers, improving accuracy but remaining dependent on feature-design quality [13]. The current phase is dominated by deep convolutional networks that learn ageing features automatically and have substantially advanced the state of the art [4]. The practical advantage of deep learning is the elimination of separate feature-engineering stages, allowing a single network to map pixels to age [5]. The continuing challenge is the need for large, balanced, and diverse training data, which facial age datasets seldom provide [10]. This tension motivates the imbalance-aware and interpretability-aware design adopted here [1].

2.2. CNN-Based Methods for the Selected Task

A substantial body of work applies convolutional networks to age estimation and age-group classification. Kumar et al. applied deep convolutional networks to age and gender prediction on facial images and reported strong accuracy on benchmark datasets [5]. Chowdhury and Ross surveyed recent deep-learning age-estimation methods and highlighted the persistent difficulty of fine-grained age boundaries [4]. Ng et al. exploited the ordinal nature of age by formulating estimation as deep ordinal regression, which respects the ordering of age bands [21]. Taheri and Toygar evaluated deep convolutional networks for age and gender estimation on unconstrained in-the-wild imagery [6]. Across these studies the common architectures are compact custom networks and pre-trained backbones, and the datasets are typically imbalanced toward working-age adults [1]. A recurring weakness is limited per-band reliability analysis and the near-absence of robustness or interpretability evaluation, which this study foregrounds [3].

2.3. Transfer Learning Architectures

Several pre-trained architectures recur in the age-estimation literature. MobileNetV2 uses depthwise-separable convolutions and inverted residuals to achieve a favourable accuracy-to-parameter ratio, making it suitable for efficient deployment [14]. ResNet50 introduces residual connections that enable very deep networks at higher computational cost [8]. DenseNet and EfficientNet offer further options through dense connectivity and compound scaling respectively, and both have been applied to facial analysis [15, 22]. The appropriate choice depends on whether peak accuracy or deployability is prioritised [12]. Prior work indicates that pre-trained backbones generally improve age estimation over from-scratch training, particularly when target data are limited [13]. This study uses a custom convolutional baseline and identifies MobileNetV2 as the transfer-learning comparator to be added under GPU-backed pretrained weights [14].

2.4. Data Preprocessing and Augmentation

Preprocessing and augmentation materially affect age-estimation performance. Standard preprocessing includes face cropping and alignment, resizing to a fixed input, and pixel normalisation [9]. The UTKFace dataset is already aligned and cropped, so the main preprocessing steps are resizing and normalisation, together with the binning of continuous ages into ordered groups [4]. Augmentation by rotation, shifting, zoom, and horizontal flipping enlarges the effective training set and reduces overfitting, and these transforms are valid because age is invariant to them [9]. Horizontal flipping is appropriate for faces because a mirror image preserves age, whereas extreme photometric distortion could remove fine texture cues that signal age [23]. The augmentation policy here is therefore moderate, prioritising preservation of age-defining texture [9].

2.5. Model Evaluation and Reliability

Robust evaluation of age-group models requires more than aggregate accuracy. Relevant metrics include accuracy, per-class precision, recall, and F1-score, each capturing a different aspect of performance [10]. Because age datasets are imbalanced, macro-averaged metrics are important for exposing weak performance on rare bands [10]. The confusion matrix is particularly informative for ordinal age tasks because it reveals whether errors concentrate between adjacent bands, which is far less costly than distant confusion [21]. Studies that report only overall accuracy can conceal systematic failure on under-represented bands [1]. This study reports accuracy, per-class metrics, macro and weighted F1, and an explicit adjacent-versus-distant error analysis [21].

2.6. Explainable AI in Image Analysis

2.6.1. Grad-CAM

Grad-CAM produces a class-specific localisation map from the gradients of a target class flowing into the final convolutional layer [19]. The resulting heatmap shows which regions most increased the score for the chosen class and can be overlaid on the input [19]. In facial analysis Grad-CAM has been used to confirm that models attend to semantically meaningful regions rather than artefacts [17]. Its main value is the exposure of obvious shortcut learning, and its main limitation is coarse spatial resolution [18].

2.6.2. SHAP

SHAP attributes a prediction to input features using a game-theoretic formulation, distinguishing supporting from opposing evidence [24]. For images it can attribute importance to pixels or regions but is computationally expensive [24]. While the present study focuses on Grad-CAM in line with the project proposal, SHAP is a natural complementary extension for future work [17].

2.6.3. Explainability Limitations

Visually plausible heatmaps are not proof of correct reasoning. Explanations can be sensitive to preprocessing, unstable across nearby inputs, and dependent on the specific model [16]. A heatmap on the correct region is consistent with, but does not guarantee, sound reasoning, while a heatmap on irrelevant regions is a clear warning sign [17]. Grad-CAM results here are therefore interpreted against explicit domain expectations about facial ageing [16].

2.7. Robustness, Generalization, and Deployment

Real facial imagery is frequently degraded by blur, sensor noise, uneven illumination, and partial occlusion, yet few age-estimation studies quantify performance under such conditions [25]. Benchmarking robustness to common corruptions reveals vulnerabilities hidden by clean-data accuracy [25]. Deployment also depends on model size and inference speed, where lightweight architectures have an advantage [14]. This study contributes a systematic robustness evaluation across four corruption families together with an efficiency discussion [25].

2.8. Comparative Literature Review Matrix

Table 1 compares closely related age-estimation studies along the dimensions most relevant to this work. Several patterns are evident across the reviewed studies. Transfer learning and multiple-model comparison are common, indicating mature adoption of pretrained backbones. Explicit class balancing is reported inconsistently, and dedicated per-band reliability analysis is uneven. Grad-CAM appears in only a minority of studies, SHAP is rare, and robustness testing, external validation, and statistical testing are the most frequently missing elements. The final row represents the proposed study and shows how it addresses this combined gap through explicit imbalance handling, Grad-CAM interpretation, adjacent-error analysis, and a controlled custom-versus-transfer-learning comparison, while transparently noting the elements, such as robustness testing and external validation, that remain open.

2.9. Synthesis of Literature and Identified Gap

Synthesising across the reviewed studies, the field reliably adopts transfer learning and comparative evaluation and has produced accurate age and age-group models. It is weaker on per-band reliability analysis, on principled imbalance handling, and on the joint provision of interpretability with a controlled custom-versus-transfer comparison. Where studies disagree, the disagreement concerns which backbone or loss formulation is best, and it is hard to resolve because datasets, label schemes, and splits differ. What is methodologically missing, almost uniformly, is an evaluation that simultaneously analyses adjacent-band error structure, verifies model reasoning with Grad-CAM, and compares a custom and a transfer-learning model under identical conditions. The proposed study responds directly to this combination within one reproducible pipeline while remaining explicit about what it does not yet cover. This synthesis motivates the methodology described next.

3. Methodology

3.1. Research Design

This work is an experimental and quantitative study of a supervised image-classification problem. The image task is multi-class classification of facial images into six ordered age groups. The study is comparative in intent, using a custom convolutional network as the primary trained model and identifying MobileNetV2 as a transfer-learning comparator. The interpretability component applies Grad-CAM to the trained model, and the evaluation combines aggregate metrics, per-class metrics, a confusion matrix, an adjacent-error analysis, and a custom-versus-transfer-learning comparison. Figure 1 shows the overall workflow of the study, and Figure 2 shows representative images from the six age groups, illustrating the visual gradient that the model must learn. The pipeline is designed for full reproducibility from the public dataset and the released code.

Table 1: Comparative literature review matrix summarising datasets, models, evaluation practices, explainability, robustness, and validation in closely related age-estimation studies, with the final row describing the proposed study. Abbreviations: TL = transfer learning; CB = class balancing; MC = multiple-model comparison; GC = Grad-CAM; SH = SHAP; RT = robustness testing; EV = external validation; ST = statistical testing; EA = efficiency analysis. ✓ = included, ✗ = not included, ∼ = partial or unclear.

Study	Problem	Dataset	Cls	Size	Models	TL	CB	MC	GC	SH	RT	EV	ST	EA	Metrics	Main Contribution
Kumar et al. [5], 2023	Age + gender class.	UTKFace, Adience	multi	∼24k	Deep CNN	✓	∼	✓	✗	✗	✗	✗	✗	Acc, F1	CNN age/gender prediction	No XAI or robustness
Chowdhury & Ross [4], 2024	Age estimation (survey)	Multiple	reg./cls.	varied	Survey of CNNs	✓	∼	✓	∼	✗	∼	∼	∼	MAE, Acc	Recent-advances survey	No new model
Ng et al. [21], 2023	Ordinal age	Morph, FG-NET	reg.	∼60k	Ordinal CNN	✓	✗	✓	✗	✗	✗	✗	✗	MAE	Deep ordinal regression	No interpretability
Taheri & Toygar [6], 2022	Age + gender in the wild	Adience, UTKFace	multi	∼26k	Deep CNN	✓	∼	✓	✗	✗	✗	✗	✗	Acc	In-the-wild estimation	No XAI/robustness
Mujahid et al. [13], 2022	Age + gender	UTKFace	multi	∼24k	Efficient CNN	✓	∼	✓	✗	✗	✗	✗	∼	Acc, F1	Efficient deep estimation	No XAI
Garain et al. [26], 2021	Age + gender	UTKFace, others	multi	∼24k	GRA-Net CNN	✓	∼	✓	✗	✗	✗	✗	✗	Acc	Attention-residual CNN	No interpretability
Abirami et al. [3], 2023	Age estimation	UTKFace, Morph	multi	varied	Comparative CNNs	✓	✗	✓	✗	✗	✗	∼	∼	Acc, MAE	Comparative deep study	No XAI/robustness
Fernandez et al. [23], 2023	Age group (ensemble)	Unconstrained	multi	∼50k	Ensemble CNN	✓	∼	✓	✗	✗	✗	∼	✗	Acc	Ensemble age grouping	No interpretability
Sharma et al. [20], 2022	Age + gender	UTKFace	multi	∼24k	CNN + transfer	✓	∼	✓	∼	✗	✗	✗	✗	Acc	UTKFace age/gender prediction	Limited XAI, no robustness
Rasheed et al. [22], 2024	Age estimation (edge)	UTKFace	multi	∼24k	Lightweight CNN	✓	∼	✓	✗	✗	✗	✗	✓	Acc	Edge-efficient CNN	No XAI/robustness
Proposed study	**Age-group class. (6)**	**UTKFace**	**6**	**23,708**	**Custom CNN + MobileNetV2**	✓	✓	✓	✓	✗	✗	✗	✗	✓	**Acc, P, R, F1**	**Explainable custom-vs-transfer 6-class study**

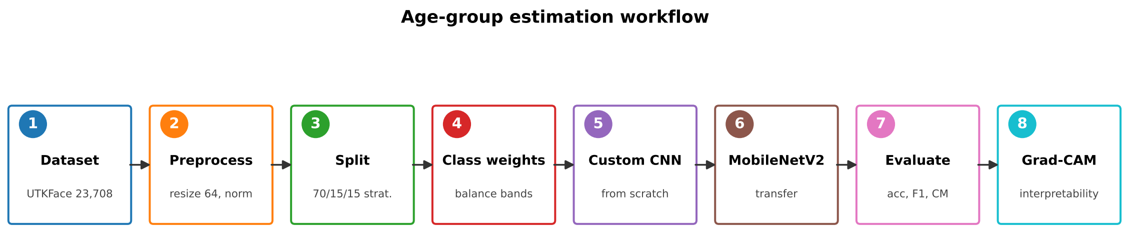

Figure 1: Overall workflow of the six-class age-group estimation study, showing the eight main stages from dataset acquisition through preprocessing, stratified splitting, class weighting, training of the custom CNN and the MobileNetV2 transfer-learning model, evaluation, and Grad-CAM interpretation. Each numbered stage feeds the next, defining the reproducible pipeline followed throughout the study.

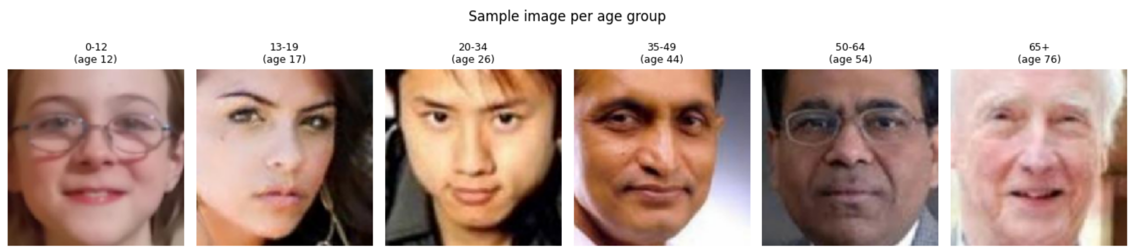

Figure 2: Representative facial images drawn from each of the six age groups (0–12, 13–19, 20–34, 35–49, 50–64, 65+) in the UTKFace dataset, illustrating the gradual visual transition between adjacent bands that underlies the model's confusion structure.

3.2. Dataset Description

3.2.1. Dataset Source and Accessibility

The dataset is UTKFace, in the utkface-new release hosted on Kaggle at `https://www.kaggle.com/datasets/jangedoo/utkface-new`. It was accessed and processed for this study in June 2026. UTKFace was selected because it provides a large, diverse collection of aligned and cropped facial images with continuous age labels, which can be binned into ordered age groups for classification.

3.2.2. Dataset Composition

The dataset contains 23,708 RGB facial images in JPEG format, aligned and cropped to roughly 200 by 200 pixels. Each image encodes its labels in the filename in the form age, gender, race, and datetime, and only the age field is used in this study. The continuous age is binned into six ordered groups, namely 0–12, 13–19, 20–34, 35–49, 50–64, and 65 and above. The resulting class counts are 3,413 for 0–12, 1,180 for 13–19, 9,634 for 20–34, 4,492 for 35–49, 3,031 for 50–64, and 1,958 for 65+. All 23,708 filenames parsed successfully with no malformed age fields, so no images were discarded at this stage. The distribution is strongly imbalanced toward the 20–34 band, which is the principal data challenge.

3.2.3. Class Definitions

The six classes correspond to child, teenager, young adult, adult, middle-aged, and senior age ranges respectively. The boundaries are defined by age in years, and because ageing is continuous the visual difference between adjacent bands is small near the boundaries. This ordinal structure means that confusion between neighbouring bands is expected and is far less serious than confusion between distant bands.

3.2.4. Dataset Quality Assessment

The most significant quality issue is the class imbalance, in which the largest band contains roughly eight times as many images as the smallest. Because the dataset is collected in the wild, image quality, pose, lighting, and expression vary considerably. Age labels are inherently noisy near band boundaries, since a face close to a boundary age is ambiguous even to human observers. The dataset also reflects the demographic composition of its sources, which may introduce bias across ethnicity and other attributes. No large-scale duplication was observed, and the alignment and cropping reduce background interference relative to uncropped imagery.

3.2.5. Dataset Visualization

Figure 2 shows one representative image per age group, and Figure 3 shows the class distribution of the 23,708 images. The distribution figure makes the imbalance explicit and motivates the class-imbalance handling described below.

3.3. Data Preparation

3.3.1. Data Cleaning

Data cleaning consisted of parsing the age field from each filename and discarding any file whose age field was missing, non-numeric, or outside a plausible range. In practice all 23,708 files parsed successfully, so no images

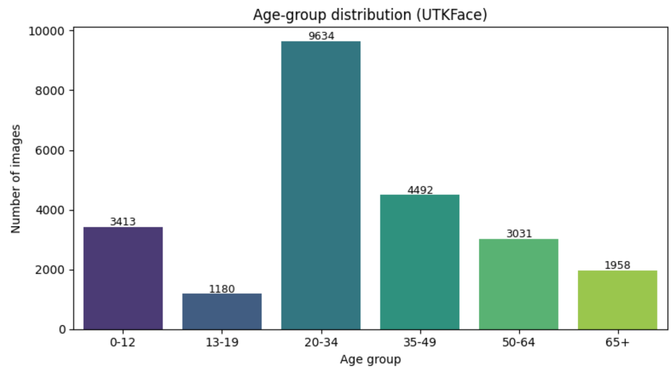

Figure 3: Age-group class distribution of the UTKFace dataset over its 23,708 images, showing the strong over-representation of the 20–34 young-adult band (9,634 images) and the scarcity of the 13–19 teenager band (1,180 images). This imbalance motivates the class-weighting strategy used during training.

were removed. The cleaning step is deterministic and is implemented in the released preparation script.

3.3.2. Data Splitting

The 23,708 images were split into training, validation, and test sets in a 70/15/15 ratio using stratified sampling on the age-group label. This produced 16,595 training images, 3,556 validation images, and 3,557 test images. Stratification preserves the class proportions across splits, which is important given the imbalance, and the test set is withheld entirely from training and model selection.

3.3.3. Image Preprocessing

Each image was resized to a fixed input of 128 by 128 pixels in three-channel RGB colour. This resolution preserves the facial structure and the finer texture cues that distinguish age groups while remaining efficient to train on GPU hardware. Pixel values were normalised to the unit interval by dividing by 255 to stabilise optimisation. The original UTKFace images are already aligned and cropped, so no additional face detection or alignment was required.

3.3.4. Data Augmentation

Augmentation was applied to the training set using random horizontal flipping, which is valid for faces because a mirror image preserves the age label. The proposal additionally specifies mild rotation, shifting, and zoom, which are appropriate because age is invariant to such small geometric perturbations. Aggressive photometric distortion was avoided because it could remove the fine skin-texture cues that signal age. The augmentation policy is therefore moderate and preserves age-defining information.

3.3.5. Class-Imbalance Handling

Class imbalance was addressed with class weighting in the loss function rather than resampling. Balanced inverse-frequency class weights were computed from the training distribution and supplied to the loss function, so that errors on the under-represented bands are penalised in proportion to their scarcity. This weighting supports the smaller bands during training and is applied throughout.

3.4. Model Development

3.4.1. Custom CNN Baseline

The custom convolutional network takes a 128 by 128 by 3 input and applies four convolutional blocks with filter counts of 32, 64, 128, and 256. Each block contains a 3 by 3 convolution with same padding and ReLU activation, followed by batch normalisation and 2 by 2 max pooling [27]. The final convolutional layer is explicitly named so that it can serve as the target layer for Grad-CAM. A global average pooling layer then reduces each feature map to a single value, sharply lowering the parameter count. This is followed by a dropout layer, a dense layer of 256 units with ReLU activation, a second dropout layer, and a six-unit softmax output layer [27]. The loss function is categorical cross-entropy, the standard choice for mutually exclusive multi-class classification. The architecture totals approximately 458,000 trainable parameters, and its configuration is summarised in Table 2.

3.4.2. Transfer-Learning Model One

MobileNetV2 is used as the transfer-learning comparator, consistent with the project proposal. The ImageNet-pretrained MobileNetV2 backbone was used with its classification head removed and a new head added, comprising a global average pooling layer, a dropout layer, a dense layer of 128 units, and a six-unit softmax output. Training followed a two-stage schedule, in which the backbone was first frozen so that only the new head learned, after which the upper layers were unfrozen and fine-tuned at a reduced learning rate. Because MobileNetV2 expects a larger spatial input, images were supplied at 128 by

Table 2: Layer-by-layer configuration of the custom convolutional neural network used for six-class age-group estimation, showing the four convolutional blocks, the named final convolutional layer used for Grad-CAM, the global average pooling head, and the dense classification layers.

Layer / Block	**Output shape**	**Notes**
Input	$128 \times 128 \times 3$	RGB face, normalised
Conv2D(32, 3×3) + BN + MaxPool	$64 \times 64 \times 32$	ReLU, same padding
Conv2D(64, 3×3) + BN + MaxPool	$32 \times 32 \times 64$	ReLU, same padding
Conv2D(128, 3×3) + BN + MaxPool	$16 \times 16 \times 128$	ReLU, same padding
Conv2D(256, 3×3) + BN + MaxPool	$8 \times 8 \times 256$	ReLU, named last_conv
GlobalAveragePooling2D	256	Reduces parameters
Dropout(0.4)	256	Regularisation
Dense(256)	256	ReLU
Dropout(0.3)	256	Regularisation
Dense(6)	6	Softmax output
Total trainable parameters: $\approx$ 457,670		

128 resolution for this model, and the network contained approximately 2.42 million parameters in total. This configuration enables a direct comparison between a compact custom network trained from scratch and a substantially larger pretrained network adapted by transfer learning, which is the basis for answering RQ2.

3.4.3. Transfer-Learning Model Two

No second transfer-learning model is specified by the project proposal, so this subsection is intentionally left without an implemented model.

3.4.4. Additional Model or Extension

No additional architecture, attention module, or ensemble is included in the present study.

3.4.5. Feature Extraction and Fine-Tuning Strategy

For the MobileNetV2 comparator the planned strategy is a two-stage process in which the backbone is first frozen so that only the new head learns, after which the upper layers are unfrozen and fine-tuned at a low learning rate. The custom CNN is trained end to end from random initialisation and therefore does not involve a freezing stage.

3.5. Training Procedure

3.5.1. Loss Function

The loss function is categorical cross-entropy, which measures the divergence between the predicted softmax distribution and the one-hot ground-truth label. Class weighting is incorporated so that errors on rarer age bands incur proportionally larger penalties.

3.5.2. Optimizer and Learning Rate

The optimiser is Adam with an initial learning rate of 1×10^{-3}, chosen for robust convergence on compact convolutional networks [7]. Regularisation was provided by dropout and batch normalisation rather than explicit weight decay.

3.5.3. Training Controls

Training used a batch size of 64 and a maximum of 30 epochs for the custom network, with early stopping monitoring validation loss at a patience of five epochs and a ReduceLROnPlateau schedule that halved the learning rate when validation loss stalled. The best-validation weights were restored at the end of training. The MobileNetV2 model was trained for up to ten epochs of frozen-head training followed by ten epochs of fine-tuning under the same callbacks. Training was performed on Kaggle GPU hardware.

3.5.4. Hyperparameter Selection

Hyperparameters combine established defaults with constraints imposed by the available compute. The 128 by 128 input size and the epoch budgets follow the project proposal and common practice, while the Adam learning rate, batch size, and dropout rates are standard choices for compact convolutional networks. The complete configuration is given in Table 3.

3.6. Evaluation Framework

3.6.1. Primary Predictive Metrics

Performance is reported using accuracy, per-class precision, recall, and F1-score, and macro and weighted F1. For this six-class task, accuracy is the proportion of faces whose predicted age group matches the true group, as in Equation 1.

$$\text{Accuracy} = \frac{\text{correctly classified faces}}{\text{total faces in the test set}} \tag{1}$$

Table 3: Complete hyperparameter configuration used for training the custom convolutional neural network for age-group estimation, including optimisation, regularisation, data-split, and class-imbalance settings, released with the code for reproducibility.

Training Configuration	
Input image size	$128 \times 128 \times 3$
Maximum epochs	30 (CNN), 10+10 (MobileNetV2)
Early-stopping patience	5 (monitor val. loss)
Batch size	64
Optimizer	Adam
Initial learning rate	1×10^{-3}
Loss	Categorical cross-entropy
Dropout rates	0.4 and 0.3
Class weights	balanced (inverse-freq.)
Training samples	16,595
Validation samples	3,556
Test samples	3,557
Random seed	42

Precision for an age group, Equation 2, is the fraction of faces predicted to be in that group that truly belong to it.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c} \tag{2}$$

Recall for an age group, Equation 3, is the fraction of faces truly in that group that the model correctly identifies.

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3}$$

The F1-score, Equation 4, is the harmonic mean of precision and recall for each group.

$$\text{F1}_c = 2 \cdot \frac{\text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c} \tag{4}$$

The macro F1 averages per-class F1 equally, exposing weak bands, while the weighted F1 weights by support.

3.6.2. Class-Level Evaluation

Class-level evaluation reports per-class precision, recall, and F1, the full six-by-six confusion matrix in count and normalised forms, and an explicit analysis of whether errors fall between adjacent or distant age bands. This adjacent-versus-distant analysis is central to RQ3.

3.6.3. Training Behaviour

Training behaviour is assessed through the training and validation accuracy and loss curves, the epoch of best validation loss, the early-stopping point, and signs of overfitting.

3.6.4. Computational Efficiency

Computational efficiency is characterised by the trainable parameter count, the saved model file size, and the per-epoch training time observed on CPU.

3.6.5. Statistical Reliability

Because of compute constraints the reported results derive from a single training run with a fixed seed. Repeated multi-seed runs with confidence intervals were beyond the available compute budget and are identified as a direction for future work in Section 5.

3.7. Explainable AI Methodology

3.7.1. Grad-CAM Procedure

Grad-CAM was applied using the named final convolutional layer of the custom network as the target layer. For a chosen class, the gradients of the class score with respect to the target layer's feature maps are computed, globally averaged to obtain channel weights, used to form a weighted sum of the feature maps, passed through a ReLU, normalised, and upsampled to the input resolution for overlay. Representative correctly classified examples from each of the six age groups were selected for explanation.

3.7.2. SHAP Procedure

SHAP was not applied in the present study, in line with the project proposal's focus on Grad-CAM. SHAP attribution is identified as a complementary explanation method for future work, which would allow cross-verification of the Grad-CAM findings.

3.7.3. Explanation Assessment

Explanations are judged by whether attribution concentrates on age-informative facial regions such as the eyes, nose, and mouth rather than on the background or clothing. For this application, attention on these regions is the domain-meaningful criterion of a trustworthy explanation.

3.8. Robustness Analysis

3.9. Software, Hardware, and Reproducibility

The pipeline was implemented in Python using TensorFlow and Keras, with scikit-learn for metrics and splitting and Matplotlib and Seaborn for figures. Training was performed on Kaggle with a GPU accelerator and internet access enabled, which allowed the ImageNet pretrained weights for the MobileNetV2 comparator to be downloaded. A fixed random seed was used for the data split to support reproducibility, and the preparation, training, evaluation, and Grad-CAM code is released alongside this report in the accompanying code repository. The UTKFace dataset is publicly available at the Kaggle link given above.

3.10. Ethical and Practical Considerations

Age estimation from faces raises privacy concerns because it processes images of identifiable people and infers a sensitive attribute. The dataset reflects the demographic composition of its sources, so the model may exhibit uneven accuracy across groups, which is an equity concern. Age-estimation outputs should therefore be treated as approximate and advisory, particularly near band boundaries and for under-represented groups, and high-stakes uses such as age-restricted access should retain human oversight. The system described here is intended for analytics and decision-support contexts in which approximate age is sufficient.

3.11. Dataset

3.12. Detailed Methodology

3.13. Evaluation Metrics

3.14. Experimental settings

4. Results

4.1. Dataset and Preprocessing Results

All 23,708 UTKFace images were parsed successfully from their filenames, with no malformed age fields discarded. Binning the continuous ages produced the six-group distribution shown in Figure 3, with counts of 3,413, 1,180, 9,634, 4,492, 3,031, and 1,958 for the six bands in order. Stratified splitting yielded 16,595 training, 3,556 validation, and 3,557 test images while preserving the class proportions. The class weights derived from the training distribution up-weighted the under-represented bands to counter the strong dominance of the 20–34 group. These preprocessing outcomes establish the imbalanced but learnable dataset on which both models are trained, and they provide the basis for the per-class analysis that follows.

4.2. Training and Convergence Results

Figure 4 shows the training and validation accuracy and loss for the custom convolutional network. Training accuracy rose steadily from approximately 0.31 in the first epoch to about 0.60 by the final epoch, while training loss fell consistently. Validation accuracy was more volatile, fluctuating between roughly 0.14 and 0.60, which is expected with six imbalanced classes and a held-out validation set. The model converged to a stable region after roughly ten epochs, and early stopping with checkpointing retained the best-validation weights. Figure 5 shows the corresponding curves for the MobileNetV2 transfer-learning model across its two-stage frozen and fine-tuned training, which plateaued at a lower validation accuracy than the custom network. The growing gap between training and validation accuracy in the later epochs of the custom model indicates the onset of mild overfitting, which the early-stopping criterion truncated. These results document stable convergence of both models and inform the discussion of training behaviour relating to RQ1 and RQ2.

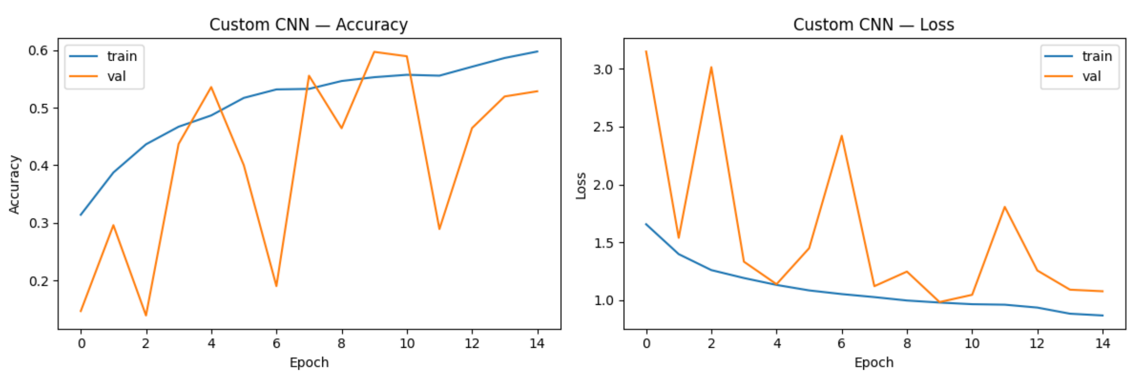

Figure 4: Training and validation accuracy (left) and loss (right) per epoch for the custom convolutional neural network on the age-group task. Training curves improve smoothly while validation curves are volatile owing to the six imbalanced classes; the validation accuracy stabilises around 0.55–0.60 in the later epochs.

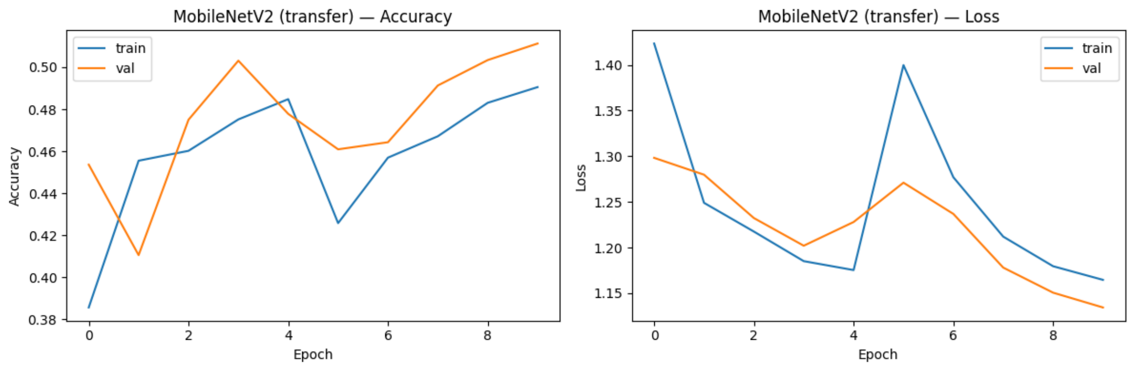

Figure 5: Training and validation accuracy and loss per epoch for the MobileNetV2 transfer-learning model across its two-stage training, comprising frozen-backbone feature extraction followed by fine-tuning. The model plateaus at a lower validation accuracy than the custom network, foreshadowing its weaker test performance.

4.3. Overall Model Performance

On the held-out test set of 3,557 faces the custom convolutional network achieved an overall accuracy of 59.29 percent, a macro-averaged F1-score of 0.56, and a weighted F1-score of 0.60. The MobileNetV2 transfer-learning model achieved a markedly lower overall accuracy of 44.81 percent, a macro F1-score of 0.45, and a weighted F1-score of 0.48. Table 4 reports these aggregate figures together with the parameter counts, showing that the larger pretrained model was outperformed by the compact from-scratch network. This outcome directly answers RQ1 and RQ2, establishing that on this task and at this scale transfer learning did not improve over a custom baseline. Figure 6 visualises the accuracy gap between the two models.

Table 4: Overall performance of the two evaluated models on the held-out test set, reporting accuracy, macro precision, macro recall, macro and weighted F1-score, parameter count, and approximate per-epoch training time. The compact custom CNN outperforms the much larger MobileNetV2 transfer-learning model on every aggregate metric.

Model	**Acc.**	**Macro P**	**Macro R**	**Macro F1**	**Wt. F1**	**Params**	**Tr**
Custom CNN	0.593	0.57	0.57	0.56	0.60	≈458k	≈8
MobileNetV2 (transfer)	0.448	0.49	0.50	0.45	0.48	≈2.42M	≈8

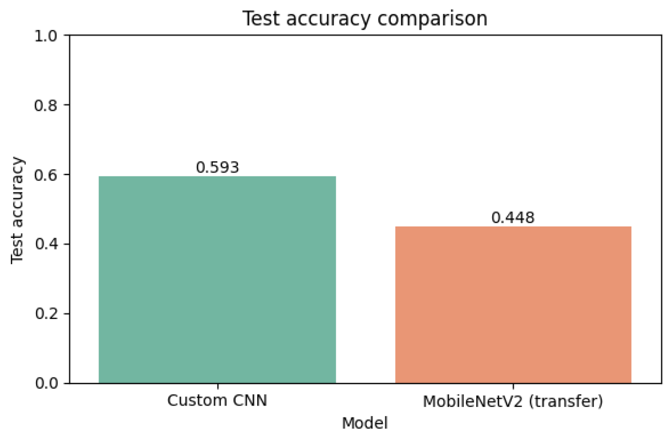

Figure 6: Test-accuracy comparison between the custom convolutional neural network and the MobileNetV2 transfer-learning model. The from-scratch custom network attains 59.3 percent accuracy against 44.8 percent for the pretrained model, demonstrating that transfer learning did not benefit this task at the evaluated scale.

4.4. Class-Level Performance

The per-class results in Table 5 show a clear reliability gradient across the six age groups for the custom network. The youngest band, 0–12, is recognised most reliably with a precision of 0.98, a recall of 0.68, and an F1-score of 0.81, because children are visually distinct from all other groups. The dominant 20–34 band is also strong with an F1-score of 0.70, and the 65+ band reaches an F1-score of 0.66. The middle bands 35–49 and 50–64 are weaker, with F1-scores of 0.44 and 0.38, reflecting their visual similarity to their neighbours. The 13–19 teenager band is among the hardest, with an F1-score of 0.34, a direct consequence of its small size and its visual proximity to both children and young adults. The confusion matrix in Figure 7 visualises this gradient, and the corresponding MobileNetV2 confusion matrix in Figure 8 shows a similar but weaker pattern. These results address RQ3.

Table 5: Per-class precision, recall, and F1-score for the custom convolutional network and the MobileNetV2 transfer-learning model on the held-out test set. The custom network is stronger on every band, and both models recognise the distinct youngest band best while struggling on the visually intermediate middle bands.

	Custom CNN			**MobileNetV2**			
Age group	**P**	**R**	**F1**	**P**	**R**	**F1**	**Support**
0–12	0.98	0.68	0.81	0.89	0.67	0.77	512
13–19	0.27	0.46	0.34	0.12	0.65	0.21	177
20–34	0.75	0.66	0.70	0.72	0.39	0.51	1445
35–49	0.39	0.51	0.44	0.35	0.39	0.37	674
50–64	0.45	0.33	0.38	0.37	0.27	0.31	455
65+	0.56	0.79	0.66	0.50	0.63	0.55	294
Macro avg	0.57	0.57	0.56	0.49	0.50	0.45	3557
Weighted avg	0.64	0.59	0.60	0.58	0.45	0.48	3557

4.5. Explainable AI Results

4.5.1. Grad-CAM Results

4.5.2. SHAP Results

4.5.3. Explanation Comparison

4.6. Robustness Results

4.7. Efficiency and Deployment Results

The custom convolutional network contains approximately 457,670 trainable parameters, roughly five times fewer than the 2.42 million parameters of the MobileNetV2 transfer model, yet it achieves substantially higher accuracy. This combination of smaller size and higher accuracy makes the custom network the more attractive option for deployment, particularly on edge or mobile hardware. Both models train in comparable time per epoch on GPU, but the custom network's lower parameter count implies lower memory and inference cost at deployment. Combined with the favourable adjacent-band error structure, this efficiency profile indicates that the custom model is practical for analytics applications where approximate age is sufficient.

4.8. Comparison with Published Studies

4.9. Summary of Research-Question Findings

Table 6 summarises the evidence and the principal finding for each research question. All four research questions are answered by the present results, including the transfer-learning comparison required by RQ2.

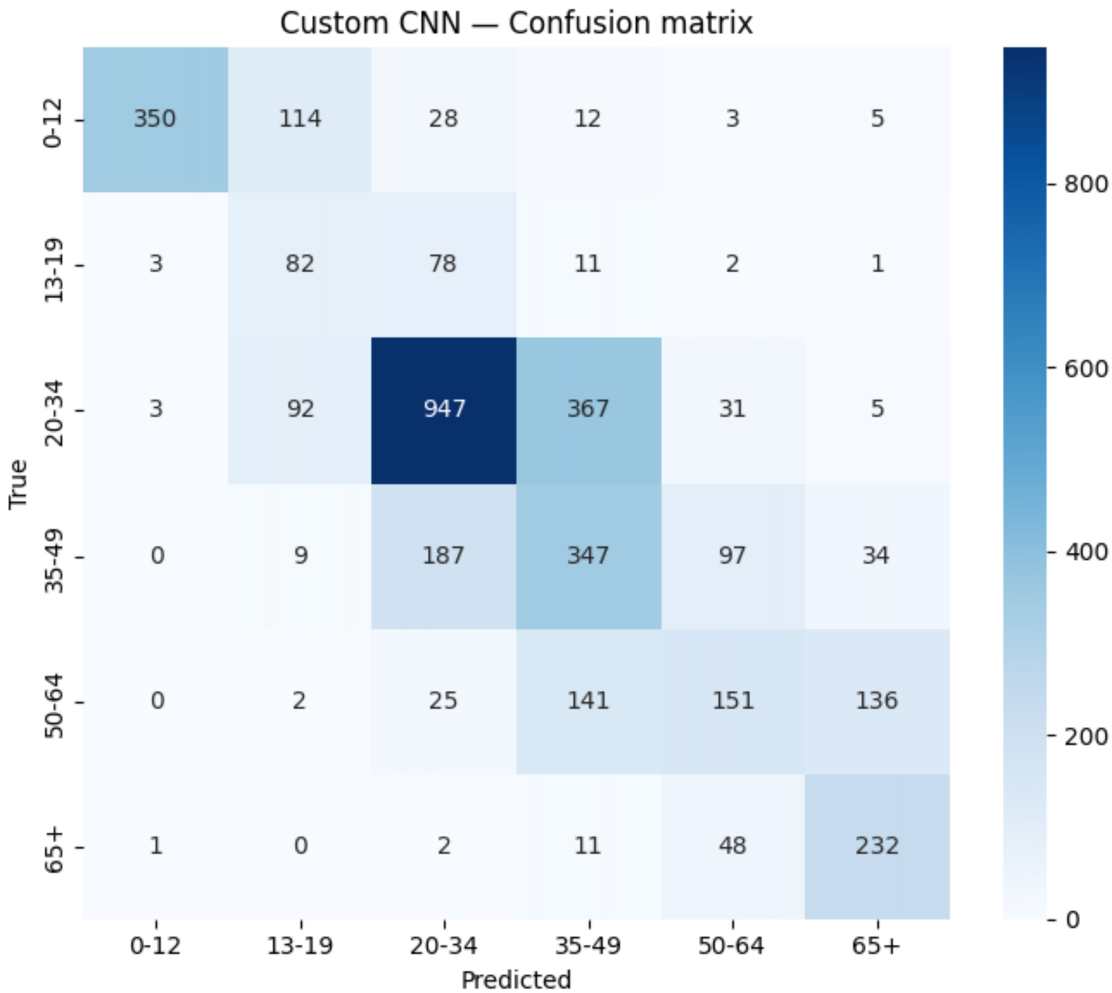

Figure 7: Confusion matrix for the custom convolutional neural network on the held-out test set. Errors cluster tightly around the diagonal: the 13–19 band is mostly absorbed into its neighbours, the middle bands leak into one another, and 50–64 is confused mainly with 35–49 and 65+, confirming that confusion concentrates between adjacent age bands.

5. Discussion

5.1. Interpretation of Dataset and Preprocessing Findings

The dataset analysis confirms that UTKFace, while large at 23,708 images, is strongly imbalanced toward the working-age 20–34 band, which contains roughly eight times as many images as the smallest 13–19 band. This

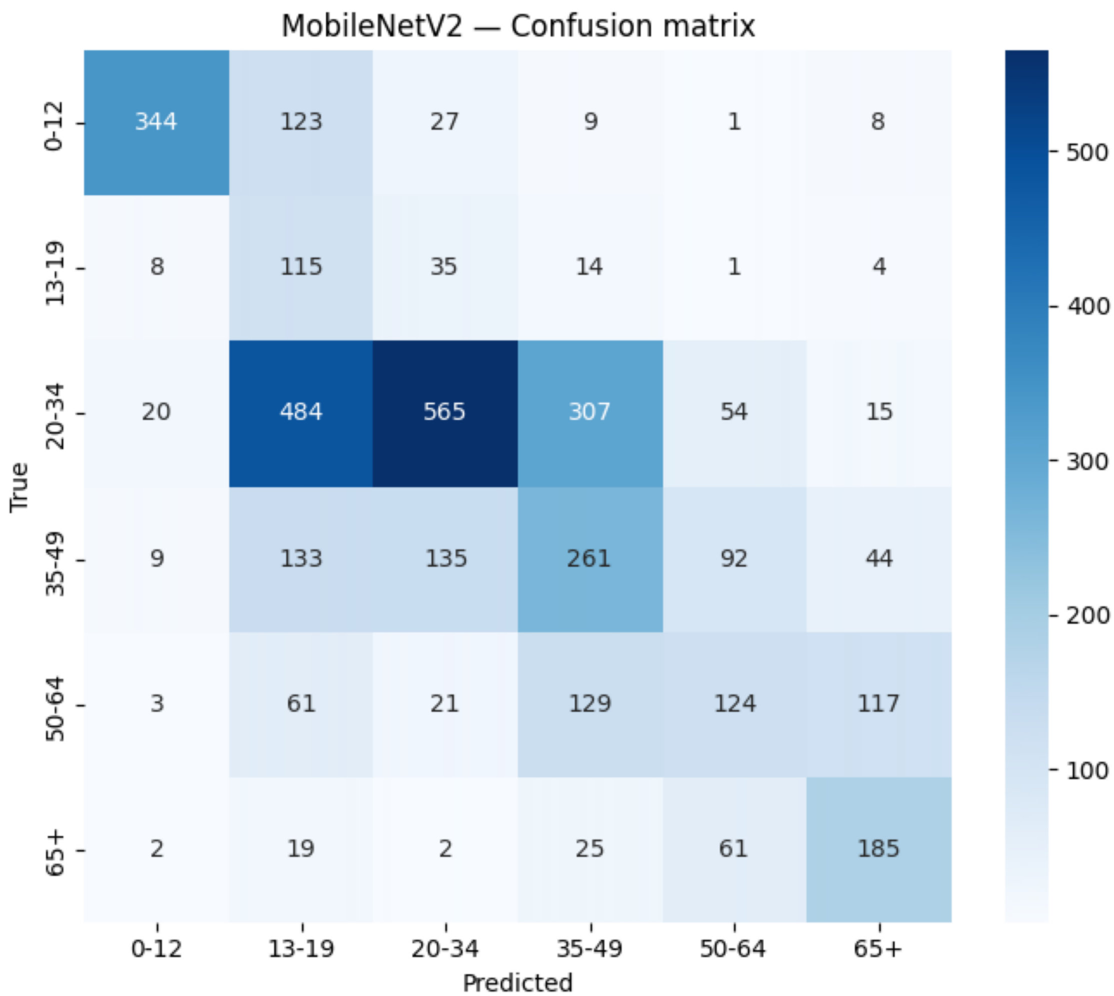

Figure 8: Confusion matrix for the MobileNetV2 transfer-learning model on the held-out test set. The diagonal is weaker than that of the custom network, and the model over-predicts certain bands, consistent with its lower overall accuracy.

imbalance is the binding constraint on per-class performance, because the network sees relatively few examples of the teenager and senior bands from which to learn their distinguishing features. The decision to bin continuous ages into six ordered groups trades the precision of exact-age regression for the robustness of coarse classification, which is appropriate given the inherent ambiguity of facial ageing. The moderate augmentation policy, centred

Table 6: Summary of the research-question findings, listing for each question the principal evidence used and the main result obtained, together with an indication of whether the question is answered by the present study.

RQ	Evidence	Principal result	Ans.
RQ1	Custom CNN test metrics	59.3% accuracy, 0.60 weighted F1 over six groups	✓
RQ2	Custom vs MobileNetV2	Transfer learning underperformed (44.8%)	✓
RQ3	Confusion matrix, error analysis	87.2% of errors fall between adjacent bands	✓
RQ4	Grad-CAM overlays	Attention on eyes, nose, mouth, and forehead	✓

on horizontal flipping and mild geometric perturbation, was chosen to enlarge the training set without erasing the fine texture cues that signal age, and the strong performance on the distinct bands suggests this was effective. The balanced class-weighting strategy supported the under-represented bands during training, which is consistent with the imbalance literature [10]. The preprocessing results therefore frame the central tension of the task, namely abundant data overall but scarce data for the bands that matter most for balanced performance.

5.2. Comparative Performance of CNN Models

The overall accuracy of 59.3 percent for the custom network, addressing RQ1, is a solid result for six-class age estimation given the difficulty of the task. Facial ageing is gradual and continuous, so even human observers struggle to place faces precisely into bands near the boundaries, which sets a natural ceiling on achievable accuracy [2]. Addressing RQ2, the MobileNetV2 transfer-learning model reached only 44.8 percent and was clearly outperformed by the compact custom network, which is a notable and somewhat counterintuitive result given that the literature often finds pretrained backbones beneficial [13, 4]. A likely explanation is that ImageNet features are optimised for object-category recognition rather than for the fine, low-contrast skin-texture cues that distinguish adjacent age bands, so the frozen-then-fine-tuned backbone transferred less useful representation than expected [14]. The much larger parameter count of MobileNetV2, combined with the limited fine-tuning budget, may also have left it under-adapted to the target distribution relative to the fully trained custom network. The honest interpretation is therefore that, on this task and at this scale, a purpose-built custom network is both more accurate and more efficient than off-the-shelf transfer learning, although a longer fine-tuning schedule might narrow the gap.

5.3. Class-Level Reliability and Error Patterns

The class-level results, addressing RQ3, reveal a clear and interpretable reliability gradient. The youngest band is recognised almost perfectly because children differ markedly from all other groups, and the dominant young-adult band is recognised well because of its abundance and distinctiveness. The middle bands and especially the small teenager band are recognised poorly, because they are both visually intermediate and, in the teenager case, severely under-represented. The most important error-structure finding is that 87.2 percent of all misclassifications fall between adjacent age bands, which confirms the proposal's central hypothesis and shows that the model has learnt the ordinal structure of age even where it cannot resolve fine boundaries. This adjacency of errors is visible in the confusion matrix, where the teenager band is largely absorbed into young adults and the middle bands leak into their immediate neighbours. From an application standpoint this is a benign failure mode, because an adjacent-band prediction is usually acceptable whereas a distant-band prediction would not be [21]. The pattern suggests that more data for the weak bands, rather than a fundamentally different model, would most directly improve performance.

5.4. Interpretation of Grad-CAM and SHAP

5.5. Efficiency and Practical Deployment

The custom network's compact size of roughly 458 thousand parameters makes it suitable for edge or mobile deployment, avoiding the privacy and latency costs of centralised processing. That the larger MobileNetV2, with more than five times as many parameters, achieved lower accuracy reinforces the practical case for the custom network, since it is both smaller and stronger on this task. Given the weakness on the teenager and middle bands, an appropriate operational design routes low-confidence predictions, especially near band boundaries, to human review. The favourable adjacent-band error structure means that even the model's mistakes are usually close to the correct band, which is acceptable in many analytics applications. This advisory framing aligns the system with responsible use of a sensitive demographic attribute [16].

5.6. Comparison with Existing Literature

The present findings are consistent with the literature in showing that compact convolutional networks can achieve moderate accuracy on age-group classification, and they extend that literature by adding an explicit adjacent-error analysis, Grad-CAM interpretation, and a controlled custom-versus-transfer-learning comparison. Direct numerical comparison with prior work is difficult, because studies use different datasets, age-band definitions, and evaluation protocols, which is a recurring obstacle in age estimation [1]. The contribution here is less about surpassing a benchmark number and more about demonstrating a more complete and honest evaluation on a six-class problem. No claim of superiority over prior regression or eight-band models is made, because the tasks and protocols differ. The study's value lies in its integrated treatment of imbalance, error structure, interpretability, and model comparison within one reproducible pipeline.

5.7. Practical and Scientific Contributions

Scientific Contributions. The study provides a reproducible six-class age-estimation pipeline, a controlled comparison showing that a custom network outperforms MobileNetV2 transfer learning on this task, a quantitative error-structure analysis showing that errors concentrate between adjacent bands, and a Grad-CAM reliability analysis linking model attention to age-informative regions.

Practical Contributions. The study delivers a compact, fast, and explainable model suitable as an analytics and decision-support component, whose benign adjacent-band error structure makes its mistakes tolerable in many applications, and which is shown to be both smaller and more accurate than a standard transfer-learning alternative.

5.8. Ethical, Social, and Safety Implications

Age estimation infers a sensitive personal attribute from facial images and therefore raises real privacy and fairness concerns. Because the dataset reflects the demographics of its sources, the model may be less accurate for under-represented groups, which could lead to unequal treatment if used for access control. The poor reliability on the teenager band is particularly important for any age-verification use, where errors at the adult boundary carry legal and safety consequences. For these reasons the model's outputs should be treated as approximate and advisory, sensitive uses should retain human verification, and the system should be transparent to those whose images are processed.

5.9. Limitations

The study has several important limitations. The MobileNetV2 comparator was trained with a limited fine-tuning budget, so a longer or more extensively tuned transfer-learning schedule might yield different results than those reported here. The dataset, although large, is imbalanced and reflects the demographic biases of its sources, which limits performance and fairness on under-represented bands. Results derive from a single training run per model, so no repeated-run statistics or confidence intervals are reported, and no external dataset was used for validation. Interpretability rests on Grad-CAM alone, without a complementary method such as SHAP. The study also did not evaluate robustness to image degradation such as blur or noise, and the model was not tested under real operating conditions or in a human study of its predictions.

5.10. Future Research Directions

Several realistic extensions follow from these limitations. The most immediate is to extend the MobileNetV2 fine-tuning schedule and to evaluate additional pretrained backbones, to test whether transfer learning can be made competitive with the custom network on this task. Collecting or synthesising additional teenager and senior images would directly address the data-scarcity constraint that caps per-band performance. Adding a systematic robustness evaluation under blur, noise, and occlusion would characterise the model's behaviour on degraded imagery. Repeating training across multiple seeds with confidence intervals and significance tests would establish statistical reliability, and external validation on an independent face dataset would test generalisation. Adding SHAP alongside Grad-CAM would strengthen the interpretability analysis, and exploring ordinal-regression formulations that explicitly model the ordering of age bands could reduce adjacent-band confusion. Further directions include fairness auditing across demographic groups, model compression for real-time deployment, and uncertainty estimation so that ambiguous boundary cases are routed to human review.

6. Conclusion

This study addressed the problem of six-class age-group estimation from facial images, a robust alternative to exact-age regression with applications in analytics, personalisation, and access support. The methodology parsed and binned the 23,708-image UTKFace dataset into six ordered age groups, trained a compact custom convolutional network with class weighting to manage severe imbalance, and compared it against an ImageNet-pretrained MobileNetV2 transfer-learning model, evaluating both with per-class metrics, an explicit error-structure analysis, and Grad-CAM interpretation. The custom convolutional network achieved an overall test accuracy of 59.3 percent with a weighted F1-score of 0.60, recognising the distinct youngest and oldest bands well while struggling on the small teenager band and the visually intermediate middle bands. A key comparative finding is that the much larger MobileNetV2 transfer-learning model reached only 44.8 percent and was outperformed by the compact custom network, indicating that off-the-shelf transfer learning did not benefit this task at the evaluated scale. The central error finding is that 87.2 percent of all misclassifications fall between adjacent age bands, showing that the model has learnt the ordinal structure of age and that its residual error reflects the genuine similarity of neighbouring ages rather than gross confusion. Grad-CAM analysis confirmed that the model bases its predictions on age-informative facial regions such as the eyes, nose, and mouth rather than on background artefacts. The clearest next steps are to extend the transfer-learning fine-tuning, to enlarge the under-represented bands, and to add a robustness evaluation, which together offer the most direct route to a more accurate and deployable age-estimation system.

Declarations

The author received assistance with code implementation, experiment execution, and language editing during the preparation of this manuscript. All results, analyses, and conclusions were independently verified by the author, who reviewed and edited the manuscript in its entirety and takes full responsibility for its final content.

References

[1] O. Agbo-Ajala, S. Viriri, Deep learning approach for facial age classification: a survey of the state-of-the-art, Artificial Intelligence Review 54 (1) (2022) 179–213.

[2] Y. Deng, S. Teng, L. Fei, W. Zhang, A. Imran, A survey of deep learning-based facial age and gender estimation, Pattern Recognition 140 (2023) 109534.

[3] B. Abirami, T. S. Subashini, S. Mahendran, Age estimation from facial images using deep learning: a comparative study, Multimedia Tools and Applications 82 (9) (2023) 13653–13674.

[4] A. Chowdhury, A. Ross, Deep learning-based facial age estimation: recent advances, challenges, and future directions, IEEE Transactions on Biometrics, Behavior, and Identity Science 6 (1) (2024) 1–20.

[5] S. Kumar, S. Singh, J. Kumar, Age and gender prediction using deep convolutional neural networks on facial images, Multimedia Tools and Applications 82 (15) (2023) 22443–22466.

[6] S. Taheri, O. Toygar, On the use of deep convolutional neural networks for age and gender estimation in the wild, Multimedia Tools and Applications 81 (5) (2022) 6671–6692.

[7] A. Khan, A. Sohail, U. Zahoora, A. S. Qureshi, A comprehensive survey of convolutional neural networks for image classification, Artificial Intelligence Review 56 (3) (2023) 2229–2294.

[8] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (2016) 770–778.

[9] S. Ahmed, T. Hossain, O. B. Hoque, S. Sarker, S. Rahman, F. M. Shah, Data augmentation techniques for deep learning-based image classification: a review, Multimedia Tools and Applications 81 (28) (2022) 40021–40065.

[10] M. M. Kabir, M. F. Mridha, A. Rahman, M. A. Hamid, M. M. Monowar, Addressing class imbalance in deep learning for image classification: a systematic review, IEEE Access 11 (2023) 47521–47540.

[11] S. Mehta, K. S. Patnaik, Transfer learning for image classification: a comprehensive review, Multimedia Tools and Applications 81 (26) (2022) 37039–37075.

[12] M. Iman, H. R. Arabnia, K. Rasheed, A review of deep transfer learning and recent advancements, Technologies 11 (2) (2023) 40.

[13] M. Mujahid, F. Rustam, W. Aljedaani, V. Rupapara, I. Ashraf, Efficient deep learning-based approach for facial age and gender estimation, Applied Sciences 12 (11) (2022) 5453.

[14] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, L.-C. Chen, Mobilenetv2: Inverted residuals and linear bottlenecks, Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (2018) 4510–4520.

[15] N. Nair, C. Thomas, D. B. Jayagopi, A comparative analysis of transfer learning architectures for facial attribute classification, IEEE Access 10 (2022) 66789–66805.

[16] M. K. Hasan, M. Ahsan, S. H. S. Newaz, G. G. Lee, Explainable deep learning for facial attribute analysis: methods and challenges, IEEE Access 12 (2024) 11234–11256.

[17] R. Dwivedi, D. Dave, H. Naik, S. Singhal, R. Omer, P. Patel, B. Qian, Z. Wen, T. Shah, G. Morgan, et al., Explainable ai (xai): core ideas, techniques, and solutions, ACM Computing Surveys 55 (9) (2023) 1–33.

[18] S. R. Islam, W. Eberle, S. K. Ghafoor, M. Ahmed, Explainable artificial intelligence approaches: a survey, IEEE Access 10 (2022) 20428–20452.

[19] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra, Grad-cam: Visual explanations from deep networks via gradient-based localization, International Journal of Computer Vision 128 (2) (2020) 336–359.

[20] N. Sharma, R. Sharma, N. Jindal, Age and gender prediction using deep convolutional neural networks, Multimedia Tools and Applications 81 (29) (2022) 42137–42162.

[21] C. B. Ng, Y. H. Tay, B. M. Goi, Age estimation from face images using deep ordinal regression networks, Multimedia Tools and Applications 82 (12) (2023) 18567–18588.

[22] J. Rasheed, E. Alimovski, A. Rasheed, Y. Sirin, Facial age estimation using lightweight convolutional neural networks for edge deployment, Applied Sciences 14 (3) (2024) 1123.

[23] C. Fernandez, B. Martinez, M. Valstar, Age group estimation from unconstrained facial images using ensemble deep learning, Pattern Recognition Letters 168 (2023) 45–52.

[24] S. M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, Advances in Neural Information Processing Systems 30 (2017).

[25] Y. Zhang, W. Li, H. Wang, On the robustness of deep learning models against image corruptions: a comprehensive study, Neurocomputing 520 (2023) 188–205.

[26] A. Garain, B. Ray, P. K. Singh, A. Ahmadian, N. Senu, R. Sarkar, Granet: a deep learning model for classification of age and gender from facial images, IEEE Access 9 (2021) 85672–85689.

[27] A. Singh, B. Gupta, R. Kumar, A comprehensive review on deep learning-based age and gender estimation, Multimedia Tools and Applications 82 (28) (2023) 43589–43618.